Global Collinearity-Aware Polygonizer for Polygonal Building Mapping in Remote Sensing

Fahong Zhang, Yilei Shi, *Member, IEEE*, and Xiao Xiang Zhu, *Fellow, IEEE*

***Abstract*—This article addresses the challenge of mapping polygonal buildings from remote sensing images and introduces a novel algorithm, the global collinearity-aware polygonizer (GCP). GCP, built upon an instance segmentation framework, processes binary masks produced by any instance segmentation model. The algorithm begins by collecting polylines sampled along the contours of the binary masks. These polylines undergo a refinement process using a Transformer-based regression module to ensure they accurately fit the contours of the targeted building instances. Subsequently, a collinearity-aware polygon simplification module simplifies these refined polylines and generates the final polygon representation. This module employs a dynamic programming technique to optimize an objective function that balances the simplicity and fidelity of the polygons, achieving globally optimal solutions. Furthermore, the optimized collinearity-aware objective is seamlessly integrated into network training, enhancing the cohesiveness of the entire pipeline. The effectiveness of GCP has been validated on three public benchmarks for polygonal building mapping. Further experiments reveal that applying the collinearity-aware polygon simplification module to arbitrary polylines, without prior knowledge, enhances accuracy over traditional methods such as the Douglas–Peucker (DP) algorithm. This finding underscores the broad applicability of GCP. The code for the proposed method will be made available at: https://github.com/zhu-xlab/GCP**

***Index Terms*—Building polygonal mapping, building polygonization, dynamic programming, instance segmentation.**

I. Introduction

EXTRACTING building footprints from high-resolution remote sensing images is crucial for understanding the human urbanization process [1] and serves as a foundational task for applications such as population estimation [2] and flood risk management [3]. While extensive research has focused on generating raster-format building footprints as binary masks [4], there remains a significant demand for vector formats. Vector formats offer more structured and compact representations, ideal for the construction of digital twins [5], and can provide a foundation for developing 3-D building models [6].

Building on prior research, several effective paradigms for polygonal building mapping have been introduced. One straightforward idea is the mask-based methods [7], [8], which generate initial building polygons by processing predicted binary masks from a semantic segmentation model using the marching squares [9] or contour tracing [10] algorithms. These initial polygons are then simplified through polyline simplification techniques, such as the Douglas–Peucker (DP) algorithm [11], to refine their structure and reduce complexity.

However, this pipeline encounters specific challenges. First, binary masks generated by semantic segmentation models based on convolutional neural networks (CNNs) [12] often exhibit curved or rounded edges at the corners of building polygons. This characteristic can lead to inaccuracies in the final polygon shapes when converting from binary masks to polygonal representations. Second, when simplifying an initial polygon that has excessively sampled vertices around the building's contour, there is a notable performance decay due to the ineffectiveness of the polygon simplification method.

This article builds on the concept of converting segmentation masks back into polygons and addresses the aforementioned challenges by introducing the global collinearity-aware polygonizer (GCP), which brings the following key innovations. First, inspired by the success of contour-based methods [13], [14], we incorporate a Transformer-based contour regression module to refine the initial contours extracted from binary masks, allowing them to more accurately align with the true object boundaries. Second, to address the performance degradation often observed in traditional polyline simplification techniques applied to mask contours, we introduce a collinearity-aware simplification algorithm. This method captures the piecewise linear geometric structure of buildings and effectively reduces contour complexity. The optimization is solved via dynamic programming [15], ensuring an exact and globally optimal solution. Moreover, the minimized objective can be seamlessly incorporated into the neural network as a loss function to supervise the upstream contour regression module, creating a more cohesive and effective polygon generation pipeline.

The main contributions of this article can be summarized as follows.

Received 27 March 2025; revised 8 July 2025 and 13 August 2025; accepted 9 September 2025. Date of publication 18 September 2025; date of current version 29 September 2025. This work was supported in part by German Research Foundation (DFG GZ: ZH 498/18-1) under Project 519016653, in part by German Federal Ministry for Economic Affairs and Climate Action in the framework of the "National Center of Excellence ML4Earth" under Grant 50EE2201C, in part by German Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV) based on a resolution of German Bundestag thorugh the EKAPEx Project under Grant 67KI32002B, and in part by Munich Center for Machine Learning. *(Corresponding author: Xiao Xiang Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3611869

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1) We propose a novel global collinearity-aware polyline simplification algorithm. Unlike previous approaches that provide only local or suboptimal solutions, our method optimizes an objective function that explicitly models the piecewise linear geometric structure of the polyline and consistently produces a globally optimal solution.
2) We propose a novel mask-based approach for polygonal building mapping, termed GCP, which comprises a polyline regression head and a polyline simplification module to convert building masks into vectorized polygons. In GCP, the proposed simplification algorithm is seamlessly integrated into the mask-to-polygon pipeline, with its objective function formulated as a loss. This design promotes greater cohesion and consistency across the entire pipeline, enabling more accurate and geometry-aware polygon generation.
3) We evaluate the proposed GCP on three widely used public benchmarks. The experimental results show that it achieves state-of-the-art performance.

II. Related Works

This section presents an overview of prior research relevant to our proposed method. We begin by examining a variety of approaches to polygonal building mapping. Subsequently, we discuss building mask extraction techniques, analyzing the advantages and disadvantages of two distinct pipelines: semantic segmentation and instance segmentation.

A. Polygonal Building Mapping

The purpose of polygon building mapping is to extract the building footprint from remote sensing images in polygon format. The extracted polygons should accurately fit the real-world building contours while being sufficiently simple, with a minimal number of edges. Polygonal building mapping is closely related to the task of building segmentation, which extracts building footprints as binary masks. Both methods aim to extract building content but present the results in vector and raster formats, respectively.

1) Mask-Based Methods: Building on the concept of converting building masks into building polygons, previous studies seek to establish a connection between these two formats. Notable studies include a method outlined in [16], which uses a Mask R-CNN [17] model to first extract binary building masks. These masks are then converted into simplified polygons using the DP algorithm, followed by optimization with the minimum description length (MDL) principle. Wei et al. [18] proposed a series of empirical, hand-crafted polygon refinement processes, such as removing overly sharp angles and merging parallel lines, to enhance the simplified polygon results generated by the DP algorithm. ASIP [19] proposes a method to partition the probability outputs from a semantic segmentation model into polygons. It first generates an initial polygon partition of the masks using KIPPI [20] and then leverages either a merging or a splitting operator to greedily minimize a predefined energy function which considers the fidelity and complexity of each of the polygon facet. Frame field learning (FFL) [8] begins by predicting an auxiliary frame field map to capture the orientations of each pixel relative to the building walls. The subsequent polygonization process encompasses several postprocessing steps, including contour extraction via the marching square algorithm, contour refinement using an active contour method (ACM)-like approach [21], corner-aware polyline simplification based on DP, and more.

In general, the primary focus of these mask-based methods is on converting building masks into their polygonal format. This conversion often employs empirical, local, greedy, or heuristic algorithms to simplify the contours of the building masks. In addition, these methods cannot be integrated into neural networks as a trainable module to be optimized simultaneously with semantic and instance segmentation models, which limits their effectiveness.

A core focus of the proposed GCP is to address these challenges by developing a globally optimal polygon simplification module. This module is not only robust but also seamlessly integrates into upstream networks, enabling a more cohesive polygon generation pipeline.

2) Contour-Based Methods: Unlike mask-based methods that generate initial polygon contours from mask predictions, contour-based methods [13], [14], [22] directly predict ordered building contours following an object detection pipeline. One key advantage of this approach over mask-based methods is its ability to eliminate the zigzag patterns caused by the discreteness of binary masks. However, similar to mask-based methods, a major challenge remains: simplifying the densely sampled predicted contours around building corners. A common solution presented in PolyBuilding [13] enhances point predictions by adding a classification head to each and using a threshold to filter out redundant points that do not correspond to building vertices. However, this solution treats each point independently, lacking awareness of the overall building geometry, which may result in missing or redundant vertices.

3) Connection-Based Methods: Connection-based methods [23], [24], [25] begin by predicting the position of a set of primitives such as vertices or edges of the buildings and then learn to order them to form polygons according to their pairwise connectivity. PolyWorld [23] first predicts the candidate building vertices using a semantic segmentation network, followed by a threshold-based filtering and a nonmaximum suppression (NMS) [26] operations. It then learns to predict a permutation matrix that captures the pairwise connection strength among these candidate vertices and finally generates the polygons by solving a linear assignment sum problem. Line2Poly [25] tries to improve over this line of method by taking the edges of the building, instead of vertices, as the primitive, so as to reduce the anglewise inaccuracy caused by individual vertices prediction. To form the final polygon, Line2Poly predicts an edgewise adjacent matrix to capture the connection strength between the predicted edges.

4) Sequential Modeling-Based Methods: Sequential modeling-based methods are similar to connection-based methods in terms of the identification of initial primitives. But instead of predicting the connectivity of these primitives, they directly decode the final polygon and output the

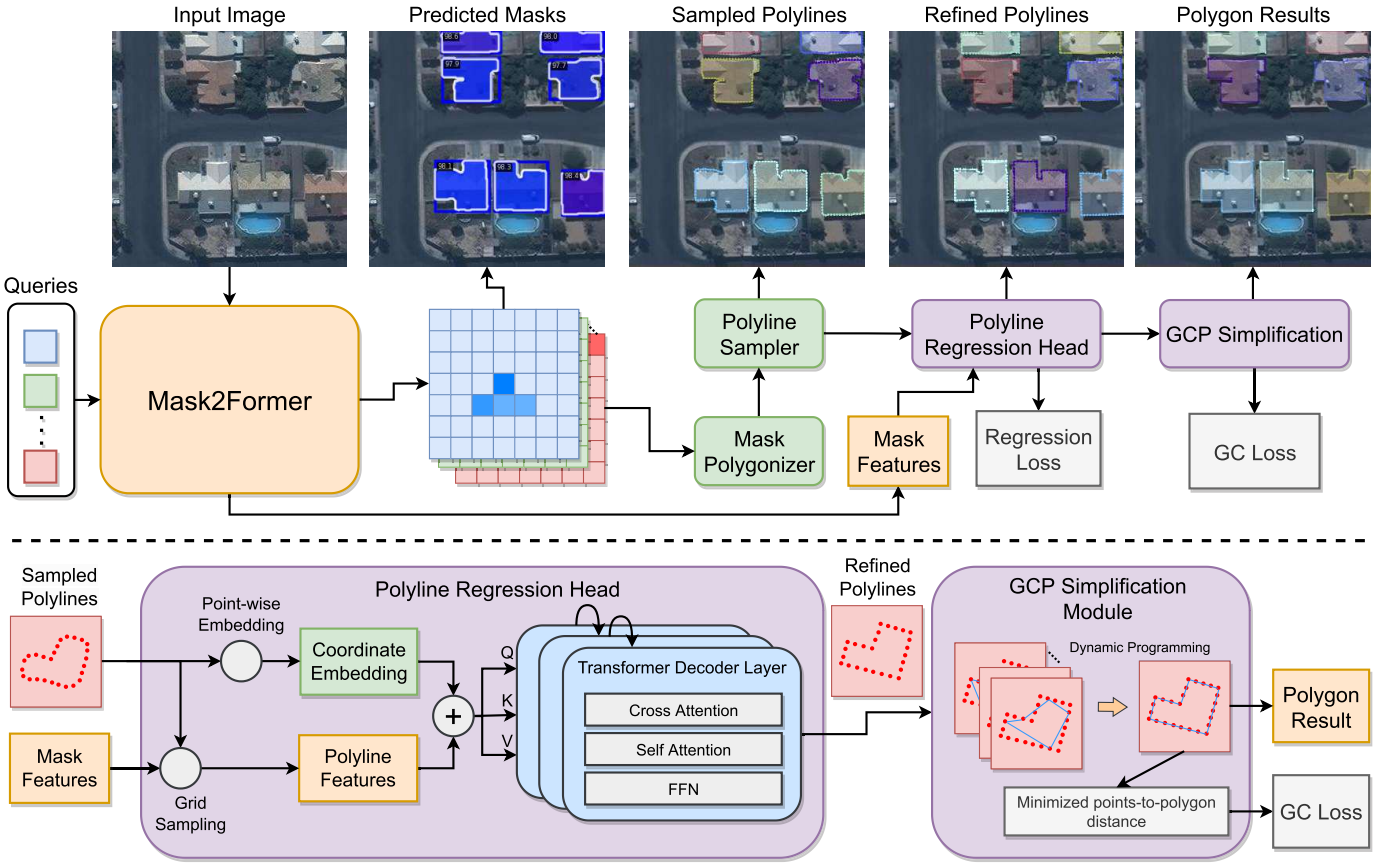


Fig. 1. Flowchart of the proposed GCP. The upper section illustrates the overall pipeline, while the bottom section provides a detailed look at the architecture of two specific modules.

final primitives step-by-step, using a sequence prediction model such as recurrent neural networks (RNNs) [27] and Transformer [28]. For example, PolyMapper [29] first generates a vertices mask for each of the building instance and then orders these vertices using an RNN. OEC-RNN [30] proposes to extract an edge attention map in addition to the vertices mask to guide the sequential decoding process conducted by ConvLSTM [31]. P2PFormer [32] introduces the utilization of various primitives, including edges and corner triplets, in addition to vertices. It incorporates a Transformer as the sequential decoder and introduces a group query mechanism. This mechanism allows the network to concentrate on different regions associated with primitives that may have multiple endpoints.

B. Building Mask Extraction

The extraction of binary building masks generally employs two distinct frameworks. The first framework utilizes semantic segmentation models to distinguish between building and non-building pixels. In scenarios that require the identification of individual building instances, it becomes essential to recognize all connected components, treating each as a separate entity. However, this approach may lead to the merging of closely situated buildings into a single structure, especially when dealing with images of lower resolution.

To address these issues, some researchers have turned to instance segmentation models, which generate separate binary masks for each building instance. A prominent approach in building extraction is Mask R-CNN [17], which extends Faster R-CNN [33] by incorporating a mask prediction head for binary segmentation within each detected bounding box. Following the introduction of Transformer [28] and DETR-based object detection frameworks [34], Transformer-based instance segmentation models such as MaskFormer [35], Mask2Former [36], and Mask-DINO [37] have emerged. These models provide simplified architectures and enhanced performance. Considering these advancements, our GCP method is built on the Mask2Former framework.

III. Methods

The workflow of the proposed GCP is illustrated in Fig. 1. The entire pipeline consists of three main stages: initial contour generation, polyline regression, and global collinearity-aware polygon simplification. These components are detailed in Sections III-A–III-C. Following this, we describe the loss functions used to train GCP in Section III-D and provide key implementation details in Section III-E.

In Section III-F, we discuss key properties of GCP and highlight its distinctions from previous methods.

A. Initial Contour Generation

The whole pipeline begins with the generation of the initial contour corresponds to the binary masks. Fig. 2 illustrates

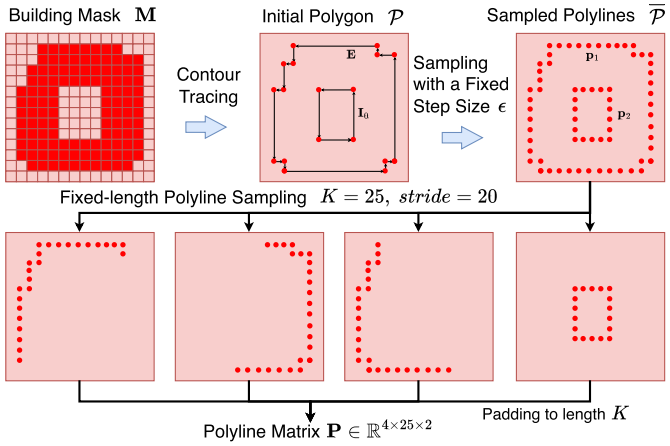


Fig. 2. Illustration of the initial contour generation process.

this process. Given an input image, the utilized Mask2Former model will generate N binary masks $\{\mathbf{M}_i\}_{i=1}^{N}$, each of which potentially corresponds to a single building instance in the image. Here, N is a predefined number of queries. Each of the predicted masks $\mathbf{M} \in \mathbb{R}^{H\times W}$ is converted to a polygon representation $\mathcal{P} = \{\mathbf{E}, \{\mathbf{I}_0, \mathbf{I}_1, \ldots\}\}$ using the contour tracing [10] algorithm, where $\mathbf{E} \in \mathbb{R}^{|\mathbf{E}|\times2}$ denotes the coordinates of the exterior polyline of $\mathcal{P}$, organized in anticlockwise order, and $|\mathbf{E}|$ is the number of points in $\mathbf{E}$. $\mathbf{E}$ is a closed polyline, ensuring $\mathbf{E}_1 = \mathbf{E}_{|\mathbf{E}|}$. $\{\mathbf{I}_0, \mathbf{I}_1, \ldots\}$ denote the interior polyines or "holes" inside $\mathbf{E}$, also closed like $\mathbf{E}$. Note that if multiple connected components exist in $\mathbf{M}$, the component with the largest area is selected to derive the polygon representation$\mathcal{P}$.

Subsequently, we uniformly sample points along each of the polyline within $\mathcal{P}$ using a fixed step size ϵ for both exterior and interior polylines and generate a set of polylines $\overline{\mathcal{P}} = \{\overline{\mathbf{p}}_1, \overline{\mathbf{p}}_2, \ldots, \overline{\mathbf{p}}_{|\overline{\mathcal{P}}|}\}$. To prevent sampled polylines from exceeding the maximum length, we set an upper limit of $L_{\max} = 512$ for each polyline. If a polyline contains more than $L_{\max}$ points, we uniformly downsample it to exactly $L_{\max}$ points to obtain $\overline{\mathbf{p}}_i$. Since each sampled polyline $\overline{\mathbf{p}}_i$ can vary in length, making them unsuitable for processing by neural networks, we apply sliding windows of size K to segment each polyline in $\overline{\mathcal{P}}$. This results in a polyline matrix $\mathbf{P} \in \mathbb{R}^{M\times K\times2}$, where M represents the total number of fixed-length polylines sampled across $\overline{\mathcal{P}}$. Polylines shorter than K are padded to match this size. Each polyline $\mathbf{P}_i \in \mathbb{R}^{K\times2}$ has a fixed length, fixed distance between two adjacent points, and can be either closed or open. In the latter case, a longer polyline from $\overline{\mathcal{P}}$ is divided into multiple segments by the sliding window, representing different segments of the original polyline.

B. Polyline Regression

Due to the above-described process, each binary mask of a building instance is represented by a polyline matrix $\mathbf{P}$ that features the contour of the mask. Such a representation is usually coarse and cannot well capture the intricate details around the building corners. To address this problem, a simple Transformer decoder is used as a polyline regression head to refine the sampled polylines in $\mathbf{P}$, as illustrated in the bottom left part of Fig. 1. The Transformer decoder comprises three layers, each mirroring the decoder configuration in Mask2Former, with both cross-attention and self-attention layers. To this end, the polyline regression head takes $\mathbf{P}$ as input and output a refined polyline matrix $\tilde{\mathbf{P}}$ according to the following equation:

$$\tilde{\mathbf{P}} = \text{RegHead}\left(\text{GridS}\left(\mathbf{P}, \mathbf{F}\right) + \text{Embed}\left(\mathbf{P}\right)\right) + \mathbf{P}. \tag{1}$$

Here, $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ denotes the features map extracted by Mask2former. GridS$(\cdot,\cdot)$ denotes the grid sampling technique, which samples the features over $\mathbf{F}$ corresponding to the coordinates of $\mathbf{P}$ using bilinear interpolation. Embed$(\cdot)$ maps any 2-D coordinates to a high-dimensional feature vector. RegHead$(\cdot)$ denotes the Transformer decoder that maps the extracted features to an offset value that refines the original polylines $\mathbf{P}$.

During the inference phase, the refined polyline matrix $\tilde{\mathbf{P}}$ is reconstructed into a closed polygon by reversing the fixed-length polyline sampling process illustrated in Fig. 2. Repetitively sampled points are merged by averaging their positions. Each reassembled polygon then undergoes a polygon simplification process introduced in Section III-C to generate the final polygon predictions.

C. Global Collinearity-Aware Polygon Simplification

Ideally, the refined polyine matrix $\tilde{\mathbf{P}}$ gives us an accurate representation of the building contours. However, these refined polylines often contain a high density of redundant points. To achieve a more simplified polygon representation, we develop a global collinearity-aware polygon simplification module.

1) Problem Formulation: Similar to the DP algorithm, the proposed polyline simplification module takes as input an arbitrary polyline

$$\mathbf{p} = (p_1, p_2, \ldots, p_T) = ((x_1, y_1), (x_2, y_2), \ldots, (x_T, y_T)).$$

Here, $T \geq 2$ is the length of the polyline, where $p_i = (x_i, y_i)$ denotes the coordinates of each point. $\mathbf{p}$ can be either closed $(p_1 = p_T)$ or open $(p_1 \neq p_T)$.

The output of the module is a *simplified polyline* $\hat{\mathbf{p}}$, which is a subsequence of $\mathbf{p}$

$$\hat{\mathbf{p}} = \left(p_{t_1}, p_{t_2}, \ldots, p_{t_m}\right) \tag{2}$$

where $(t_1, t_2, \ldots, t_m)$, satisfying $1 = t_1 < t_2 < \cdots < t_m = T$, represents a list of indices that identify the subsequence. $m \geq 2$ is the number of the chosen indexes, and it may vary for a different $\hat{\mathbf{p}}$. Note that the first and the last points of $\mathbf{p}$ are always included. Our goal is to find an optimal simplified polyline $\hat{\mathbf{p}}^*$ that captures the intrinsic geometry of $\mathbf{p}$.

2) Collinearity-Aware Objective: Our core insight in searching for the optimal $\hat{\mathbf{p}}^*$ is to maximally preserve the collinearity of consecutive point sequences within $\mathbf{p}$. We define $\mathcal{L}(\mathbf{p}, \hat{\mathbf{p}})$ as a measurement of distance

$$\mathcal{L}\left(\mathbf{p}, \hat{\mathbf{p}}\right) = \sum_{i=1}^{m-1} \sum_{j=t_i+1}^{t_{i+1}-1} \text{dis}\left(p_j, \overline{p_{t_i} p_{t_{i+1}}}\right). \tag{3}$$

Here, $\mathrm{dis}(p_j, \overline{p_{t_i} p_{t_{i+1}}})$ denotes the distance from point p_j to line $\overline{p_{t_i} p_{t_{i+1}}}$, which can be calculated by

$$\mathrm{dis}\left(p_j, \overline{p_{t_i} p_{t_{i+1}}}\right) = \frac{\left\| \overrightarrow{p_{t_i} p_{t_{i+1}}} \times \overrightarrow{p_{t_i} p_{t_j}} \right\|}{\left\| \overrightarrow{p_{t_i} p_{t_{i+1}}} \right\|} \tag{4}$$

where $\overrightarrow{p_i p_j}$ denotes the vector from point p_i to point p_j.

Intuitively, $\mathcal{L}(\mathbf{p}, \hat{\mathbf{p}})$ capture the sum of distance from each point of $\mathbf{p}$ to the segment that covers it according to $\hat{\mathbf{p}}$. A small $\mathcal{L}(\mathbf{p}, \hat{\mathbf{p}})$ indicates that most of the consecutive points in $\mathbf{p}$ that lie in the same line are well represented by the corresponding segments that are chosen in $\hat{\mathbf{p}}$, which means the inherent collinearity among points are well preserved. To this end, we define our optimization objective for searching the optimal $\hat{\mathbf{p}}^*$ as

$$\hat{\mathbf{p}}^* = \underset{\hat{\mathbf{p}}}{\operatorname{argmin}} \ \mathcal{L}(\mathbf{p}, \hat{\mathbf{p}}). \tag{5}$$

However, this definition is not well-posed, as it allows for a trivial solution: the original polyline $\mathbf{p}$ itself, since $\mathcal{L}(\mathbf{p}, \mathbf{p}) = 0$. This contradicts our intent that a "simplified polyline" should be minimalistic, ideally comprising just a few segments. To address this, we introduce a regularization term designed to penalize the inclusion of additional points in $\hat{\mathbf{p}}$

$$\hat{\mathbf{p}}^* = \underset{\hat{\mathbf{p}}}{\operatorname{argmin}} \ \mathcal{L}(\mathbf{p}, \hat{\mathbf{p}}) + \lambda |\hat{\mathbf{p}}| \tag{6}$$

where $|\hat{\mathbf{p}}| = m$ is the number of the selected points in $\hat{\mathbf{p}}$, which can vary for a different $\hat{\mathbf{p}}$. λ is a coefficient used to balance the two terms. By minimizing this objective function, we ensure that the simplified polygon $\hat{\mathbf{p}}^*$ remains compact, yet effectively captures the underlying piecewise linear geometry of $\mathbf{p}$.

3) Optimization by Dynamic Programming: Equation (6) is a combinatorial optimization problem defined in discrete space. Here, we provide an optimal solution to it using the dynamic programming technique [15]. For clarity, we first define

$$\tilde{\mathcal{L}}(\mathbf{p}, \hat{\mathbf{p}}) = \mathcal{L}(\mathbf{p}, \hat{\mathbf{p}}) + \lambda |\hat{\mathbf{p}}|. \tag{7}$$

Given two integers i and k, satisfying $1 \le i < T$ and $1 < k \le T - i$, it can be observed that the following equation holds:

$$\tilde{\mathcal{L}}\left(\mathbf{p}_{i:i+k}, \hat{\mathbf{p}}^*_{i:i+k}\right) = \min_{i<j\le i+k} \tilde{\mathcal{L}}\left(\mathbf{p}_{j:i+k}, \hat{\mathbf{p}}_{j:i+k}\right) + \sum_{l=i+1}^{j-1} \mathrm{dis}\left(p_l, \overline{p_i p_j}\right) + \lambda. \tag{8}$$

Here, $\mathbf{p}_{i:i+k} = (p_i, p_{i+1}, \ldots, p_{i+k})$ denotes a consecutive subsequence of $\mathbf{p}$, $\hat{\mathbf{p}}_{i:i+k}$ is the corresponding simplified polyline, and $\hat{\mathbf{p}}^*_{i:i+k}$ defines the specific simplification plan that yields the minimum $\tilde{\mathcal{L}}$ value. $\tilde{\mathcal{L}}(\mathbf{p}_{i:i+k}, \hat{\mathbf{p}}^*_{i:i+k})$ defines a subproblem of the origin polyline simplification problem, modifying the input polyline line $\mathbf{p}$ to $\mathbf{p}_{i:i+k}$. Specifically, to ensure (8) remains meaningful when $j = i + k$, we define

$$\tilde{\mathcal{L}}\left(\mathbf{p}_{i:i}, \hat{\mathbf{p}}^*_{i:i}\right) = 0. \tag{9}$$

Consequently, when $k = 1$, the solution is obvious

$$\tilde{\mathcal{L}}\left(\mathbf{p}_{i:i+1}, \hat{\mathbf{p}}^*_{i:i+1}\right) = \lambda. \tag{10}$$

To this end, $\tilde{\mathcal{L}}(\mathbf{p}_{i:i+2}, \hat{\mathbf{p}}_{i:i+2})$ can be resolved by combining (8) and (10). By repeating this process, we are able to solve the subproblem $\tilde{\mathcal{L}}(\mathbf{p}_{1:T}, \hat{\mathbf{p}}_{1:T})$, which is equivalent to the original problem $\tilde{\mathcal{L}}(\mathbf{p}, \hat{\mathbf{p}})$. The optimization process is illustrated in Fig. 3.

D. Loss Functions

This section presents the loss functions for supervising the training of the proposed GCP.

1) Regression Loss: During the training of Mask2Former, each predicted binary mask $\mathbf{M}$ is assigned with a ground-truth polygon $\overline{\mathbf{p}}$ via the Hungarian algorithm [38]. This ground-truth polygon can be used to supervise the predicted polylines $\tilde{\mathbf{P}}$ generated by (1). Since $\tilde{\mathbf{P}} \in \mathbb{R}^{M \times K \times 2}$ may contain multiple polylines, we randomly sample a single one $\tilde{\mathbf{p}}$ for loss calculation. Then, we calculate a Hungarian assignment between $\mathbf{p}$ and $\overline{\mathbf{p}}$ in a pointwise manner. The $\mathbf{p}$ here refers to the original polyline corresponds to $\tilde{\mathbf{p}}$ without the refinement process in (1). Only the vertex pairs with distances less than a predefined threshold $\epsilon = 15$ are selected. For these matched vertex pairs, $\tilde{\mathbf{p}}^{\text{matched}}$ and $\overline{\mathbf{p}}^{\text{matched}}$, we compute a smooth $L1$ loss [33] between them

$$\mathcal{L}_{\text{vert}} = \mathrm{SmoothL1}\left(\tilde{\mathbf{p}}^{\text{matched}}, \overline{\mathbf{p}}^{\text{matched}}\right). \tag{11}$$

In addition, we calculate an angular loss similar to the method used in PolyWorld [23]. For each set of three consecutive matched points, we calculate the angular difference and then adopt the maximum difference per polyline to determine the angular loss

$$\mathcal{L}_{\text{ang}} = \max_{(u,v,w)} \left| \tilde{\angle}_{u,v,w} - \overline{\angle}_{u,v,w} \right| \tag{12}$$

where $\tilde{\angle}_{u,v,w}$ and $\overline{\angle}_{u,v,w}$ indicate the angles formed by three consecutive points indexed by u, v, and w of $\tilde{\mathbf{p}}^{\text{matched}}$ and $\overline{\mathbf{p}}^{\text{matched}}$, respectively.

2) Global Collinearity Loss: To make the entire polyline refinement and simplification pipeline more cohesive, we incorporate supervision into the refined polylines conditioned on the final decoded polygon results. In addition, we aim to enhance the collinearity of the predicted polygons. To achieve this, we calculate a global collinearity loss for each decoded polygon. Specifically, given a refined polyline $\tilde{\mathbf{p}}$ sampled from $\tilde{\mathbf{P}}$, and its simplified result $\tilde{\mathbf{p}}^*$ obtained using (6), we define the global collinearity loss $\mathcal{L}_{\text{gc}}$ as follows:

$$\mathcal{L}_{\text{gc}} = \mathcal{L}(\tilde{\mathbf{p}}, \tilde{\mathbf{p}}^*) \tag{13}$$

which is our unnormalized objective function value conditioned on the optimal solution $\tilde{\mathbf{p}}^*$. According to (3), it can be reformulated as

$$\mathcal{L}_{\text{gc}} = \sum_{i=1}^{|\tilde{\mathbf{p}}^*|-1} \sum_{j=t_i^*+1}^{t_{i+1}^*-1} \mathrm{dis}\left(\tilde{p}_j, \overline{\tilde{p}_{t_i^*} \tilde{p}_{t_{i+1}^*}}\right) \tag{14}$$

where $\{t_i^*\}$ denotes selected indexes of $\tilde{\mathbf{p}}$ corresponds to $\tilde{\mathbf{p}}^*$. Since $\mathcal{L}_{\text{gc}}$ is essentially a sum of point-to-line distance, it remains differentiable with respect to $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{p}}^*$.

E. Implementation of GCP

This section provides key implementation details of the proposed GCP simplification algorithm. Its pseudo-code is presented in Algorithm 1. Section III-E1 analyzes its computational complexity, while Section III-E2 discusses various acceleration strategies.

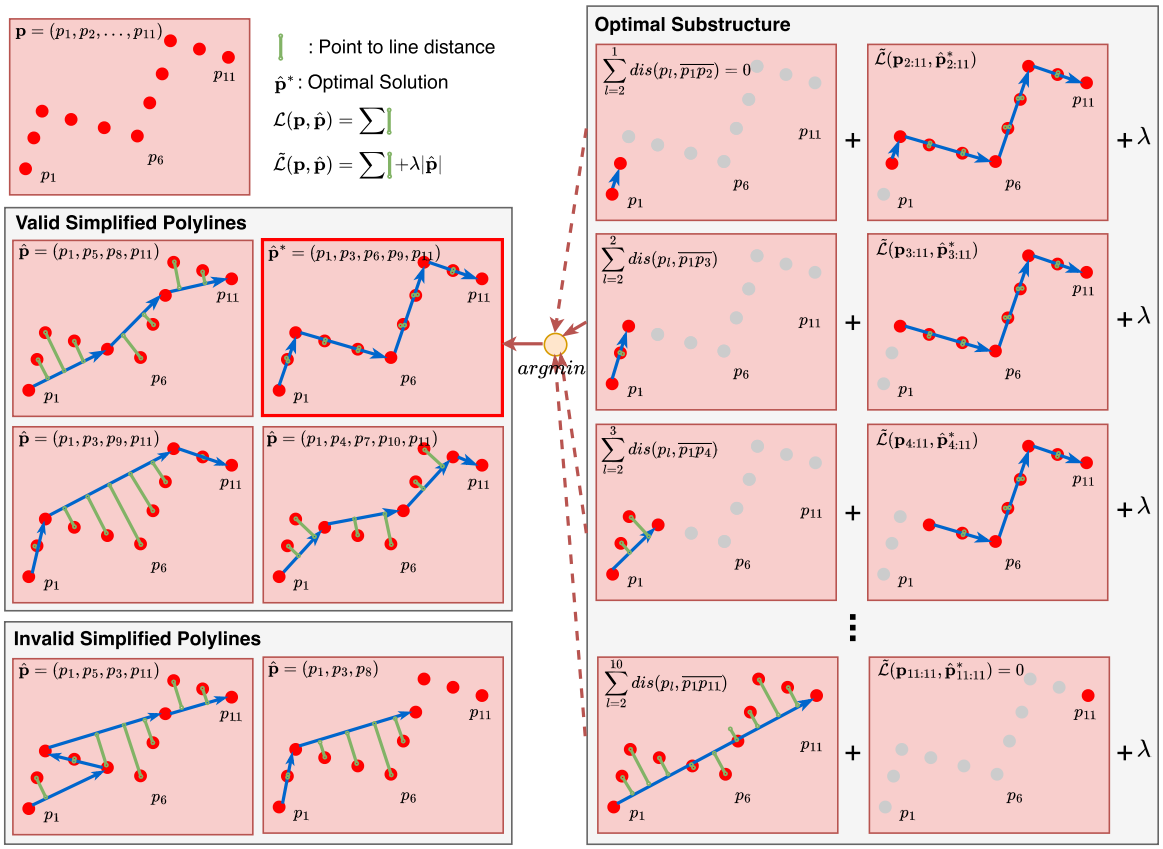


Fig. 3. Illustration of the proposed global collinearity-aware polyline simplification method. For a given polyline to be simplified $\mathbf{p}$, the middle left part of the figure lists four different possible simplification plans. The bottom left part shows two invalid simplification plans according to the definition in (2). The right part of the figure demonstrates how the optimal solution is achieved by combining the solutions to its subproblems according to (8). Such a property is termed as "optimal substructure" in dynamic programming.

1) Computational Complexity: To solve (6), we first precompute the point-to-line distances for all point–line pairs (line 17 in Algorithm 1). We then progressively solve (8) from $k = 2$ to $T - 1$ (lines 18–31 in Algorithm 1).

To improve computational efficiency, we introduce a maximum step size $k_{\max} \leq T$, which constrains the range of j in (8), ensuring $j - i \leq k_{\max}$. Intuitively, this means that the longest edge of the simplified polygon $\hat{\mathbf{p}}$ can span at most $k_{\max}$ consecutive points. This restriction reduces the time complexity of solving (6) from $O(T^3)$ to $O(T \cdot k_{\max}^2 + T^2 \cdot k_{\max})$, where $O(T \cdot k_{\max}^2)$ accounts for precomputing the point-to-line distances, and $O(T^2 \cdot k_{\max})$ corresponds to the iterative solution of (8).

2) Acceleration of GCP Simplification: Algorithm 1 presents a naive implementation of the proposed GCP simplification algorithm, which extensively utilizes loop structures to compute pairwise point-to-line distances and iteratively solve (8). However, it is important to note that most of these loops can be parallelized and vectorized, as each iteration is independent of the previous ones. Specifically, the only necessary loop structures that cannot be parallelized are the loop in line 21, which iterates over the subsequence length of $\mathbf{p}$ in (8), and the loop in line 35, which sequentially decodes the simplified polyline.

Thus, Algorithm 1 can be significantly accelerated by parallelizing and vectorizing the computation of key variables, including $\mathbf{D}$, $\tilde{\mathbf{L}}$, and $\mathbf{F}$. In addition, the GCP simplification method can be further optimized through batch processing and GPU acceleration. For more implementation details, please refer to our published codes.

F. Discussion

The proposed GCP simplification method has two major advantages compared with previous polyline simplification methods.

1) Global Optimality: The iterative optimization procedure described in (8) yields a globally optimal and exact solution to the objective function defined in (6). This offers a clear advantage over previous polygon simplification methods, such as the DP algorithm employed in [8] and [18] and the MDL principle adopted in [16], which are inherently local or yield only approximate solutions.

2) Integration With Neural Network Frameworks: The loss function $\mathcal{L}_{\text{gc}}$ in (13) is defined as the unnormalized objective function of the polyline simplification algorithm and is differentiable with respect to both the refined polyline outputs $\tilde{\mathbf{p}}$ and the simplified polylines $\tilde{\mathbf{p}}$. Although the operation for selecting $\tilde{\mathbf{p}}^*$ from $\tilde{\mathbf{p}}$ is nondifferentiable, the introduction of $\mathcal{L}_{\text{gc}}$ nonetheless establishes an effective bridge between the polyline simplification module and the upstream neural network components.

Algorithm 1 GCP Simplification Algorithm

Require: A polyline $\mathbf{p} \in \mathbb{R}^{T\times 2}$, maximun step size k_{max}, regularization coefficient λ in Eq. 6.
Ensure: Simplified polyline $\mathbf{p}^*$.

function POINT2LINE($\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$)
  Set d to the distance from point $\mathbf{p}_3$ to line $\overline{\mathbf{p}_1\mathbf{p}_2}$.
  return d
end function
function CALDISMAT($\mathbf{p}$, k_{max})
  // Get point-to-vector distance in Eq. 3.
  Initialize $\mathbf{D}$ as a zero matrix of shape (T, T).
  for $i = 1$ to T **do**
    for $j = i + 1$ to $min(T, i + k_{max})$ **do**
      for $k = i + 1$ to $j - 1$ **do**
        $\mathbf{D}_{i,j} \leftarrow \mathbf{D}_{i,j} + \text{Point2Line}(\mathbf{p}_i, \mathbf{p}_j, \mathbf{p}_k)$.
      end for
    end for
  end for
  return D
end function
$\mathbf{D} \leftarrow \text{CalDisMat}(\mathbf{p}, k_{max})$.
Define $\mathbf{\Lambda} \in \mathbb{R}^{T\times T}$ with all elements equal to λ.
$\mathbf{C} \leftarrow \mathbf{D} + \mathbf{\Lambda}$. // Cost Matrix.
Initialize $\tilde{\mathbf{L}}, \mathbf{F} \in \mathbb{R}^{T\times T}$ to zero matrices.
for $k = 1$ to T **do**
  for $i = 1$ to $T - k$ **do**
    $\tilde{\mathbf{L}}_{i,i+k} \leftarrow +\infty$.
    for $j = i + 1$ to $i + min(k, k_{max})$ **do**
      if $\tilde{\mathbf{L}}_{j,i+k} + \mathbf{C}_{i,j} < \tilde{\mathbf{L}}_{i,i+k}$ **then**
        $\tilde{\mathbf{L}}_{i,i+k} \leftarrow \tilde{\mathbf{L}}_{j,i+k} + \mathbf{C}_{i,j}$. // Eq. 8.
        $\mathbf{F}_{i,i+k} \leftarrow j$ // Record the index.
      end if
    end for
  end for
end for
Initialize $\mathbf{p}^*$ as an empty set.
$\mathbf{p}^*_1 \leftarrow 1$.
Set $j \leftarrow 1, i \leftarrow 2$.
while $j < T$ **do**
  $\mathbf{p}^*_i \leftarrow \mathbf{F}_{j,T}$.
  $j \leftarrow \mathbf{F}_{j,T}$.
  $i \leftarrow i + 1$.
end while
return $\mathbf{p}^*$

IV. EXPERIMENTS

In this section, we present and analyze the experimental results to validate the effectiveness of the proposed GCP framework.

A. Datasets and Evaluation Metrics

1) Datasets: Three widely used public benchmarks were employed in our experiments: the CrowdAI [39], WHU-Mix (vector) [22], and SpaceNet [40] datasets.

The CrowdAI dataset consists of aerial RGB images with a ground sampling distance of approximately 0.2 m. It includes a training split with 280 741 images and a validation split with 60 317 images, all sized 300 × 300 pixels. We chose this dataset to evaluate the proposed method because it is one of the most commonly used in the field of polygonal building mapping, providing numerous baseline comparisons. However, it is important to note a valid study by Adimoolam et al. [41] that highlights a potential data leakage issue in the CrowdAI dataset, which may reduce the reliability of comparative results.

The WHU-Mix (vector) dataset comprises a large collection of aerial and satellite images from various sources, including WHU [42], CrowdAI, OpenAI [43], SpaceNet [40], and Inria [44]. The dataset covers five continents and spans over ten regions worldwide, with a spatial resolution ranging from 0.09 to 2.5 m. It includes a training set of 43 778 images, a validation set of 2922 images, and two testing sets containing 11 675 and 6011 images, respectively. Notably, the second testing set does not overlap geographically with the training set, presenting a greater challenge for the generalizability of building mapping models.

The SpaceNet dataset consists of high-resolution satellite imagery paired with vector building footprint annotations. It includes data collected from multiple cities across different continents, with a ground sampling distance of approximately 0.3 m. To ensure fair comparison with prior work, we follow the experimental setup introduced in [45], where experiments are conducted on a subset of 3851 images from Las Vegas. The data are split into training, validation, and testing sets with a ratio of 8:1:1. Each image has a native resolution of 650 × 650 pixels.

2) Evaluation Metrics: We employed a variety of metrics to evaluate the effectiveness of our proposed method, including intersection over union (IoU), MS-COCO metrics, maximum tangent angle (MTA) error [8], corrected-IoU (C-IoU) [23], and the N-ratio [23].

IoU is calculated as the intersection over union area between predicted and ground-truth polygons. Following the convention in [23], the MS-COCO metrics provide average precision (AP) and average recall (AR) values for the predicted masks across IoU thresholds ranging from 0.5 to 0.95, in increments of 0.05. Specifically, we report AP and AR at IoU thresholds of 0.5 and 0.75, denoted as AP_{50}, AP_{75}, AR_{50}, and AR_{75}. In addition, to ensure comparability with the results reported in SAMPolyBuild [45], we adopt the same evaluation metrics, including mask AP (AP^{mask}) computed from raw instance masks. The metrics are reported across different object scales, such as small and medium (S&M) and large (L), in alignment with the evaluation protocol of SAMPolyBuild.

The MTA metric measures anglewise discrepancies between predicted and actual ground-truth contours. It considers contour pairs with an IoU greater than 0.5, where the predicted contour is uniformly sampled at intervals of 0.1 pixels and mapped onto the ground truth. The tangent angle differences between consecutive points on the sampled and projected sequences are calculated, with the maximum angle difference across each contour pair serving as the MTA value.

C-IoU refines the IoU calculation by scaling it according to the relative difference in the number of predicted and ground-truth vertices for each image.

TABLE I
QUANTITATIVE COMPARATIVE RESULTS ON CROWDAI DATASET

Methods	Backbone	AP↑	AP_{50}↑	AP_{75}↑	AR↑	AR_{50}↑	AR_{75}↑	IoU↑	MTA↓	C-IoU↑	N-ratio
PolyMapper [29]	VGG16	50.8	81.7	58.6	47.6	70.8	55.5	77.6	-	67.5	-
FFL (ACM Poly) [8]	UResNet101	61.3	87.4	70.6	64.9	89.4	73.9	84.1	33.5	73.7	1.13
PolyWorld [23]	R2U-Net	63.3	88.6	70.5	75.4	93.5	83.1	91.3	32.9	88.2	0.93
BuildingMapper [22]	DLA-34	63.9	90.1	75.0	-	-	-	-	-	-	-
HiT [14]	ResNet50	64.6	91.9	78.7	75.5	93.8	83.5	-	31.7	88.6	1.00
P2PFormer [32]	ResNet50	66.0	91.1	77.0	-	-	-	-	-	-	-
HiSup [7]	HRNetV2	79.4	92.7	85.3	81.5	93.1	86.7	**94.3**	-	**89.6**	-
PolyBuilding [13]	ResNet50	78.7	96.3	89.2	84.2	97.3	92.9	94.0	32.4	88.6	0.99
GCP (Ours)	ResNet50	**81.5**	**96.9**	**91.2**	**85.4**	**97.6**	**93.6**	93.4	**31.6**	84.0	0.83

Finally, the N-ratio measures the ratio of predicted vertices to ground-truth vertices. An N-ratio below 1 indicates fewer predicted vertices than the ground truth, while a ratio above 1 indicates an overprediction of vertices.

B. Implementation Details

GCP is built on the Mask2Former framework with ResNet50 [46] as the backbone. For hyperparameters, the sampling step size ϵ and the sliding window size in Fig. 2 are set to 4 and 64, respectively. The regularization coefficient λ in (6) is set to 2 for the CrowdAI and SpaceNet datasets and 4 for the WHU-Mix (vector) dataset. The maximum step size $k_{\max}$, as discussed in Section III-E1, is set to 64. All experiments are conducted on a single NVIDIA A40 GPU.

The training process for the proposed GCP model is divided into two stages. In the first stage, we train the baseline Mask2Former model. In the second stage, the parameters of the Mask2Former model are frozen, and training is focused solely on the polygonization process. At this point, only the polyline regression head and a convolutional mapping layer, which reprojects the mask features extracted by Mask2Former, remain as trainable components.

For the experiments on the CrowdAI dataset, the number of queries used to generate predicted masks with Mask2Former is set to 100. All images are resized to 320 × 320 to ensure divisibility by 32, preventing pooling issues within the network. Batch size is set to 16. The predicted polygons are then resized back to the original 300 × 300 dimensions for metric calculation. Data augmentation techniques include random horizontal and vertical flipping, as well as random rotations of 90°, 180°, and 270°. The Mask2Former model is trained for 100 epochs using the AdamW optimizer [47], with an initial learning rate of 10^{-4}, which is decayed to 10^{-5} after the 80th epoch. In the second stage, GCP is trained for 80 000 iterations (roughly equivalent to five epochs), starting with an initial learning rate of 10^{-4}, which is reduced to 10^{-5} after the 60 000th iteration.

For the experiments on the WHU-Mix (vector) dataset, the number of queries is set to 300, and all images are resized to 512 × 512 during both the training and inference stages. The batch size is set to 8. Similar to the CrowdAI setting, the predicted results are resized back to their original dimensions for evaluation. In the first stage, Mask2Former is trained for 50 epochs with an initial learning rate of 10^{-4}, which is reduced to 10^{-5} after the 40th epoch. In the second stage, GCP is trained for 12 epochs, starting with an initial learning rate of 10^{-4}, decaying to 10^{-5} after the ninth epoch.

For experiments on the SpaceNet dataset, the number of queries is set to 100, and images are resized to 1024 × 1024, following the configuration of SAMPolyBuild [45]. The batch size during training is set to 4. In the first stage, Mask2Former is trained for 100 epochs with an initial learning rate of 10^{-4}, which is reduced to 10^{-5} after 80 epochs. In the second stage, GCP is trained for 50 epochs, with learning rate decay applied after epoch 40.

Since the proposed method follows an instance segmentation pipeline, the output from Mask2Former consists of multiple binary masks, which may overlap. To accurately compute metrics other than the MS-COCO metrics, we apply a confidence threshold of 0.5 to filter out redundant masks. The remaining masks are then aggregated to generate a single mask for calculating IoU and C-IoU values. Similarly, only predicted polygons with a confidence score higher than 0.5 are considered when calculating MTA and N-ratio.

C. Experimental Results

In this section, we compare the proposed GCP method with the other state-of-the-art methods on the utilized 3 datasets and also conduct an ablation study to analyze the significance of each proposed network component. The methods used for comparison include PolyMapper [29], FFL [8], PolyWorld [23], BuildingMapper [22], HiT [14], P2PFormer [32], HiSup [7], PolyBuilding [13], Line2Poly [25], RSPrompter [48], and SAMPolyBuild [45]. The metric results of these methods are cited from their original papers, with consistent experimental settings. For visualization in Fig. 4, we generate the results of the comparison methods by running the official inference codes.

1) Results on CrowdAI Dataset: The quantitative comparative results on the CrowdAI dataset are shown in Table I. In terms of the COCO metrics, the proposed GCP achieves AP, AP_{50}, and AP_{75} values of 81.5, 96.9, and 91.2, outperforming the previous methods. In terms of IoU metrics, GCP outperforms most of the methods but is slightly outperformed by HiSup and PolyBuilding. This may be attributed to the aggregation of multiple masks predicted by Mask2Former into a single semantic mask, which can reduce performance.



Fig. 4. Visualized polygonal building predictions for different comparative methods on the CrowdAI dataset. Red circles indicate areas where the building structure is not accurately represented.

Compared with previous methods, GCP achieves a significantly lower N-ratio, indicating that it produces a more compact polygon representation than human annotations. This can be attributed to two key factors. First, the CrowdAI dataset

TABLE II
QUANTITATIVE COMPARATIVE RESULTS ON WHU-MIX (VECTOR) DATASET. †RESULTS ARE CITED FROM [32]

Test1								
Method	Backbone	AP↑	AP_{50}↑	AP_{75}↑	IoU↑	MTA↓	C-IoU↑	N-ratio
BuildingMapper† [22]	DLA-34	58.0	82.6	65.9	-	-	-	-
Line2Poly† [25]	DLA	58.6	81.9	66.2	-	-	-	-
P2PFormer [32]	ResNet50	60.6	87.3	68.9	-	-	-	-
GCP (Ours)	ResNet50	**62.0**	**87.6**	**69.1**	79.6	32.4	68.8	0.96
Test2								
Method	Backbone	AP↑	AP_{50}↑	AP_{75}↑	IoU↑	MTA↓	C-IoU↑	N-ratio
BuildingMapper† [22]	DLA-34	48.1	73.2	52.0	-	-	-	-
Line2Poly† [25]	DLA	48.9	73.3	52.8	-	-	-	-
P2PFormer [32]	ResNet50	50.7	79.9	54.4	-	-	-	-
GCP (Ours)	ResNet50	**52.9**	**80.1**	**56.5**	77.5	34.2	65.7	0.94

annotations contain consecutive duplicate vertices, introducing unnecessary complexity and redundancy. Second, by minimizing the global collinearity loss $\mathcal{L}_{\text{gc}}$, GCP encourages the generation of polygons with fewer vertices. As a result, for rounded buildings, which are typically annotated with densely placed vertices, GCP predicts significantly fewer vertices.

Since the predicted polygons have fewer vertices than the ground truth, GCP exhibits a lower C-IoU compared with some previous methods. However, we argue that GCP provides a compact and simplified polygon representation without compromising accuracy metrics such as AP. This property is highly desirable for many applications.

In Fig. 4, we visualize the predicted polygons generated by various comparative methods alongside the proposed GCP on the CrowdAI dataset. It can be observed that different methods encounter distinct challenges. For example, FFL and PolyWorld struggle with cases where building polygons contain internal holes, while HiSup occasionally produces polygons with sharp angles and redundant vertices. In addition, PolyWorld may omit crucial vertices, particularly when they are located close to another building's vertex. In contrast, the proposed GCP method effectively addresses these issues, generating simplified yet accurate polygons.

2) Results on WHU-Mix-Vector Dataset: The quantitative comparative results for the two test sets of the WHU-Mix (vector) dataset are presented in Table II. In test set 1, the proposed GCP demonstrates superior performance in terms of AP, AP_{50}, and AP_{75}. The AP_{50} and AP_{75} values for GCP are slightly better than those of the previous state-of-the-art method, P2PFormer, with margins of 0.2–0.3. Notably, GCP's AP value surpasses that of P2PFormer by a more significant margin of 1.4, suggesting that the proposed method achieves higher precision at larger IoU thresholds and is potentially better at capturing detailed building geometries.

On the more challenging test set 2, the proposed GCP outperforms the second-best method by a larger margin. Specifically, GCP's AP value is higher than that of P2PFormer by 2.2, and its AP_{75} value exceeds P2PFormer by 2.1, indicating that GCP generalizes better to unseen scenarios. This advantage may stem from GCP's mask-to-polygon pipeline, where mask predictions are usually less sensitive to domain shifts. Furthermore, the N-ratio values for GCP on both Tests 1 and 2 are below 1.0, indicating that the generated polygons are more compact than the ground-truth polygons.

3) Results on SpaceNet Dataset: The quantitative comparison results on the SpaceNet dataset are presented in Table III. Compared with other methods, GCP achieves the highest scores in AP^{poly}, AR^{poly}, and AP^{mask}, demonstrating superior performance in delineating building objects of various sizes. However, GCP shows relatively lower performance on large buildings, with reduced AP_L^{poly} and AP_L^{mask} values. This may be attributed to the point sampling strategy of the underlying Mask2Former architecture, which can result in insufficient supervision for large-scale objects. Notably, GCP exhibits the smallest gap between AP^{mask} and AP^{poly} at only 0.4, compared with SAMPolyBuild (1.7), HiSup (1.4), and FFL (8.6). This suggests that GCP is more effective at converting building masks into accurate vector representations while preserving their geometric integrity.

4) Ablation Study: We conducted some ablation study experiments on the test set 1 of the WHU-Mix (vector) dataset. As shown in Table IV, we verify the influence of the polyline regression module, the proposed global collinearity loss $\mathcal{L}$, as well as the utilized polygon simplification method.

The first row of Table IV presents the performance of the predicted binary masks. The second row reflects the performance of a naive method that converts masks to polygons using the contour tracing algorithm [10]. While the accuracy of the predicted masks is largely preserved, this approach results in a significantly high N-ratio and very low C-IoU, indicating that the output polygons contain many redundant vertices.

The third and fourth rows denote the performance when simplifying the initial contour without using the polyline regression module. Here, we observe a drop in performance for the AP and IoU metrics, but a significant improvement in C-IoU values and a substantial reduction in N-ratio, suggesting that the polygons are significantly simplified.

In the last four rows, the polyline regression module is applied, utilizing the regression losses $\mathcal{L}_{\text{vert}}$ and $\mathcal{L}_{\text{ang}}$ to supervise the module. Comparing these results to those obtained with the polyline regression module disabled, we observe a significant reduction in MTA values. This indicates an

TABLE III

QUANTITATIVE COMPARATIVE RESULTS ON SPACENET DATASET. †RESULTS ARE CITED FROM [45]

Method	Backbone	AP^{poly}	$AP^{poly}_{S\&M}$	AP^{poly}_{L}	AR^{poly}	AP^{mask}	$AP^{mask}_{S\&M}$	AP^{mask}_{L}	IoU
FFL (ACM Poly)† [8]	UResNet101	52.7	52.4	61.5	58.2	44.1	43.7	60.1	77.6
HiSup† [7]	HRNetV2	57.7	57.3	71.6	61.7	59.1	58.7	73.1	84.1
RSPrompter† [48]	Vit-B	-	-	-	-	60.5	60.4	**74.3**	84.0
SAMPolyBuild [45]	Vit-B	59.2	58.6	**72.8**	63.5	60.9	60.4	73.6	**84.3**
GCP (Ours)	ResNet50	**61.1**	**60.9**	69.8	**67.5**	**61.5**	**61.3**	70.4	82.3

TABLE IV

ABLATION STUDY ON WHU-MIX VECTORS TEST 1. "POLY." DENOTES WHETHER THE BINARY MASKS PREDICTED BY MASK2FORMER ARE CONVERTED TO POLYGONS. $\mathcal{L}_{\text{VERT}} + \mathcal{L}_{\text{ANG}}$ DENOTES WHETHER THE POLYLINE REGRESSION HEAD IS USED. $\mathcal{L}_{\text{GC}}$ DENOTES WHETHER THE GLOBAL COLLINEARITY LOSS IS EMPLOYED. "SIMP." DENOTES WHICH POLYGON SIMPLIFICATION METHOD IS UTILIZED. "DP" DENOTES THE DOUGLAS–PEUCKER ALGORITHM. "CLS" DENOTES ADDING A CLASSIFICATION HEAD TO FILTER OUT THE REDUNDANT VERTICES. "GCP" DENOTES THE PROPOSED POLYLINE SIMPLIFICATION ALGORITHM

Poly.	$\mathcal{L}_{vert} + \mathcal{L}_{ang}$	$\mathcal{L}_{gc}$	Simp.	AP↑	AP_{50}↑	AP_{75}↑	IoU↑	MTA↓	C-IoU↑	N-ratio
✗	-	-	-	**62.0**	**87.7**	**69.3**	**79.8**	-	-	-
✓	✗	✗	None	**62.0**	**87.7**	**69.3**	**79.8**	37.2	12.5	13.7
✓	✗	✗	DP	59.3	87.2	66.7	79.2	45.5	63.0	1.36
✓	✗	✗	GCP	60.9	87.6	68.6	79.6	46.4	64.5	1.24
✓	✓	✗	CLS	59.4	86.8	66.5	78.9	33.9	68.7	1.10
✓	✓	✗	DP	60.3	87.2	67.4	79.3	36.1	62.5	1.45
✓	✓	✗	GCP	60.6	87.3	67.7	79.4	33.2	68.3	0.98
✓	✓	✓	GCP	**62.0**	87.6	69.1	79.6	**32.4**	**68.8**	0.96

increased geometric similarity between the predicted polygons and the ground-truth polygons.

We also compare the effectiveness of different polygon simplification methods. In addition to the commonly used DP algorithm and the proposed GCP, we implement a simplification method referred to as "CLS," which classifies each point on the predicted contour as either a vertex or a nonvertex point. To achieve this, we add a vertex classification branch within the polyline regression head and filter out redundant points below a predefined threshold of 0.1 during the inference process.

Comparing these simplification methods, we find that "CLS" achieves lower MTA and N-ratio values than DP but results in a relatively significant performance decline in the AP, AP_{50}, and AP_{75} metrics. This may be attributed to the fact that "CLS" enables more precise localization of building vertices, producing polygons that are better aligned with the ground truth. However, it is also more susceptible to failure, as the omission of a single vertex can lead to the exclusion of a significant portion of the building, resulting in substantially lower AP scores. Among the three methods, the proposed GCP consistently outperforms the others in terms of AP and IoU metrics, regardless of whether the polyline regression module is employed. In addition, GCP achieves a lower N-ratio and MTA values, indicating that it provides more compact and simplified results while effectively capturing the structural and geometric information of the building polygons.

The impact of the proposed $\mathcal{L}_{\text{gc}}$ loss can be observed by comparing the results of the last two rows. The inclusion of $\mathcal{L}_{\text{gc}}$ significantly enhances the overall performance of the polygonization pipeline. It not only reduces the performance gap between the polygonal and mask prediction results but also yields a more compact and simplified polygon representation. This suggests that the integration of the GCP simplification module to the upstream neural networks is significant in producing more consistent and well-aligned polygon results

In Fig. 5, we present visualization results of several ablated models using different simplification methods on Tests 1 and 2 of the WHU-Mix (vector) dataset. It can be observed that when the vertex classification-based method, "CLS," is used, some true building vertices may be missing, leading to the omission of significant portions of the building (as shown in the first row). When the "DP" method is applied, certain redundant points on the building contour are not properly removed, resulting in an overly complex polygon representation. In contrast, the proposed GCP method is optimized to identify collinear relationships within the sampled building polylines, allowing it to generate a more compact and smooth polygon representation.

5) Sensitivity Analysis of λ: The only key hyperparameter in the proposed GCP approach is the λ in (6) that is used to balance the fidelity and complexity of the simplified polyline. This section verifies the sensitivity of GCP to λ under two different cases.

In the first case, we apply the proposed polyline simplification method to simplify the initial polygons sampled from the mask contours without being processed by the polyline regression module and report the metrics under different λ values. According to Table V, we can observe that when $\lambda = 0$, the simplification method deteriorates to naively keeping all the points in the initial polygon, resulting in an over complicated polygon representation. With the increase of

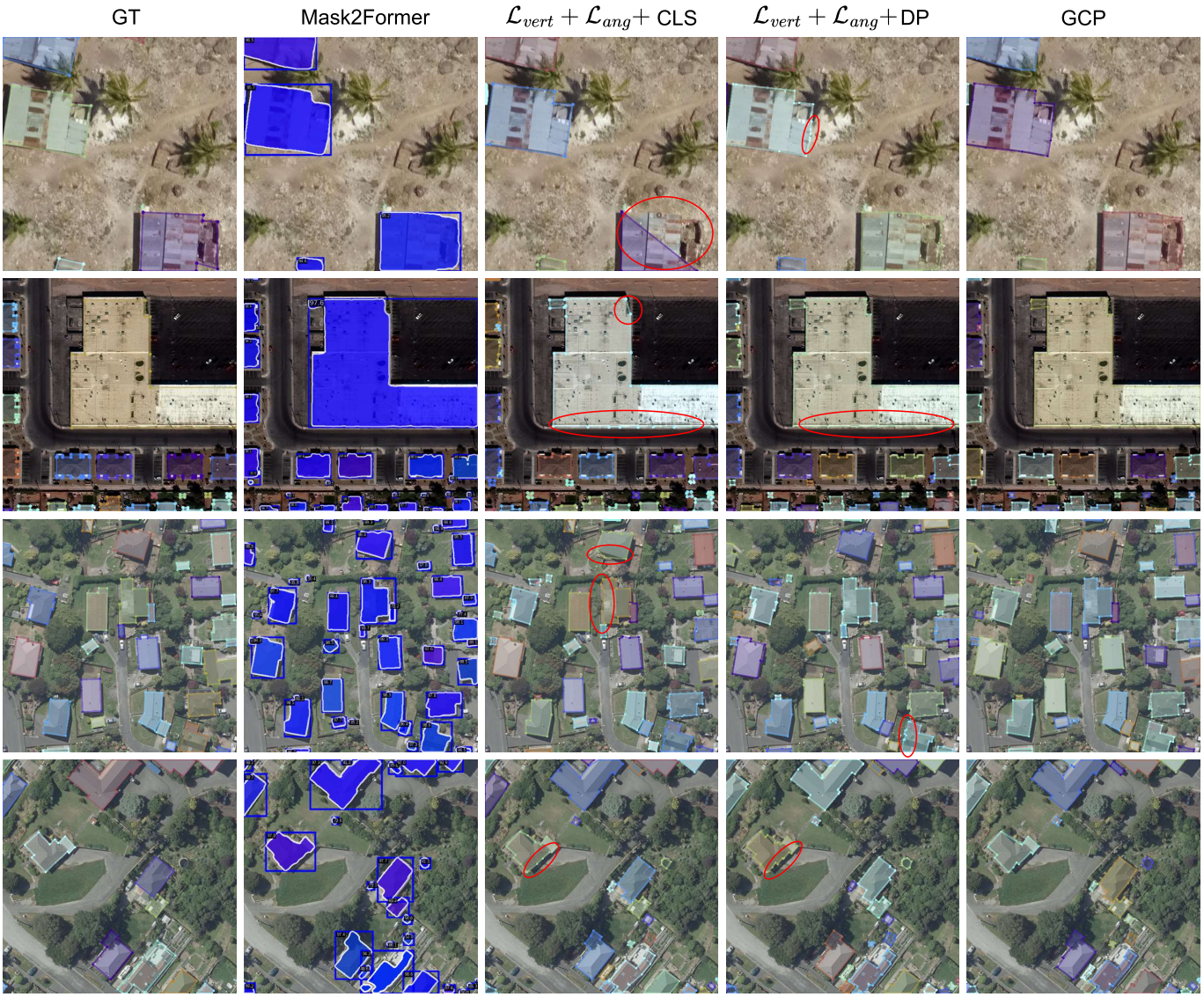


Fig. 5. Visualized polygonal building predictions for various ablated models on WHU-Mix (vector) dataset. Red circles indicate areas where the building structure is not accurately represented. The images in the first two rows pertain to the Test 1 set, whereas the last two rows feature images from the Test 2 set.



Fig. 6. Polygon simplification results of the proposed global collinearity-aware polyline simplification module under different λ values. The simplification is applied directly to initial contours, without refinement from the polyline regression head.

λ, the accuracy of the polygon gradually drops, resulting in lower AP and IoU values. In meanwhile, the complexity of the polygon results is gradually reduced, resulting in lower N-ratio and higher C-IoU values. We visualize the simplification results for different λ values in Fig. 6. As the λ value increases, the resulting polygons become progressively smoother.

In the second case, we evaluate the performance of the full GCP model under different λ values across various datasets. The quantitative and qualitative results are presented in Table VI and Fig. 7, respectively. From Table VI, we observe that varying λ impacts the simplicity of the generated polygons, as reflected in the changes of the N-ratio. However,

TABLE V

SENSITIVITY OF THE PROPOSED POLYLINE SIMPLIFICATION METHOD TO HYPERPARAMETER λ. POLYLINE REGRESSION MODULE IS DISABLED

λ	AP	AP_{50}	AP_{75}	IoU	MTA	C-IoU	N-ratio
0	62.0	87.7	69.3	79.8	37.2	12.5	13.7
1	61.7	87.7	69.2	79.7	47.5	49.0	2.06
2	61.4	87.7	69.0	79.7	47.4	58.0	1.55
4	60.9	87.6	68.6	79.6	46.4	64.5	1.24
8	59.4	87.3	66.9	79.3	44.0	67.5	1.00



Fig. 7. Visualization of polygonal building mapping results produced by GCP under different λ values across multiple datasets. The first, second, and third rows correspond to samples from the CrowdAI, WHU-Mix (vector), and SpaceNet datasets, respectively.

AP values remain relatively stable, particularly on the WHU-Mix (Vector) and SpaceNet datasets. This demonstrates that GCP can flexibly generate building polygons with varying levels of geometric simplicity to meet user pReferences, while largely preserving fidelity. Fig. 7 further shows that the shapes of the generated polygons exhibit less variation compared with those in Fig. 6. This phenomenon can be attributed to the polyline regression head, which captures the intrinsic geometry of building contours more accurately, producing a piecewise linear representation that makes GCP less sensitive to changes in λ.

6) Sensitivity of the Polyline Regression Head to Network Architecture: We investigate the sensitivity of the polyline regression head to architectural design by varying the number of Transformer decoder layers and reporting the corresponding metrics. As shown in Table VII, these changes result in only minor fluctuations (less than 0.1–0.3) across all metrics. This indicates that GCP is relatively robust to the specific architectural configuration of the polyline regression head.

TABLE VI

SENSITIVITY OF THE PROPOSED POLYLINE SIMPLIFICATION METHOD TO HYPERPARAMETER λ. POLYLINE REGRESSION MODULE IS ENABLED

CrowdAI							
λ	AP	AP_{50}	AP_{75}	IoU	MTA	C-IoU	N-ratio
1	81.3	96.8	91.0	93.3	33.4	85.8	1.04
2	81.5	96.9	91.2	93.4	31.6	84.0	0.83
4	78.9	96.7	89.1	92.6	32.8	80.1	0.75
WHU-Mix Vector test1							
λ	AP	AP_{50}	AP_{75}	IoU	MTA	C-IoU	N-ratio
1	61.5	87.7	68.6	79.5	38.8	55.2	1.75
2	61.9	87.7	69.0	79.6	34.8	65.8	1.22
4	62.0	87.6	69.1	79.6	32.4	68.8	0.96
SpaceNet							
λ	AP	AP_{50}	AP_{75}	IoU	MTA	C-IoU	N-ratio
1	61.1	88.9	71.4	82.2	39.6	64.3	1.53
2	61.1	88.9	71.4	82.3	35.6	74.4	1.08
4	61.2	89.0	71.8	82.3	33.7	74.1	0.87

TABLE VII

SENSITIVITY OF THE POLYLINE REGRESSION HEAD TO THE NETWORK ARCHITECTURE

Num Layers	AP	AP_{50}	AP_{75}	IoU	MTA	C-IoU	N-ratio
3	62.0	87.6	69.1	79.6	32.4	68.8	0.96
6	62.0	87.7	69.2	79.6	32.4	69.1	0.96
9	61.9	87.7	69.1	79.6	32.3	69.1	0.96

TABLE VIII

COMPUTATIONAL EFFICIENCY OF THE PROPOSED GCP TESTED ON CROWDAI DATASET. THE TIME CONSUMPTION OF EACH MODULE TO PROCESS A SINGLE 320×320 IMAGE IS REPORTED

Mask2Former	Poly OP	Poly Refine	Poly Simp.	Total
4.9 ms	17.4 ms	4.6 ms	5.3 ms	32.3 ms

7) Computational Efficiency: We report the time consumption of different modules of the proposed GCP on the CrowdAI dataset. In Table VIII, "Mask2Former" refers to the time required for Mask2Former to generate building masks. "Poly OP" represents polygon-related operations such as converting binary masks to initial polygons, sampling points along the polygon contour, assembling the refined polylines into closed polygons, and other similar processes. All these operations are executed on a single CPU core. "Poly Refine" indicates the time consumption of the polyline regression module, while "Poly Simp." refers to the proposed polygon simplification method.

Overall, it can be observed that mask generation, polyline refinement, and polygon simplification each consume approximately the same amount of time. Other auxiliary polygon operations take more time but can be easily accelerated using multiprocessing techniques for time-sensitive applications. For large-scale inference tasks, these polygon operations can be

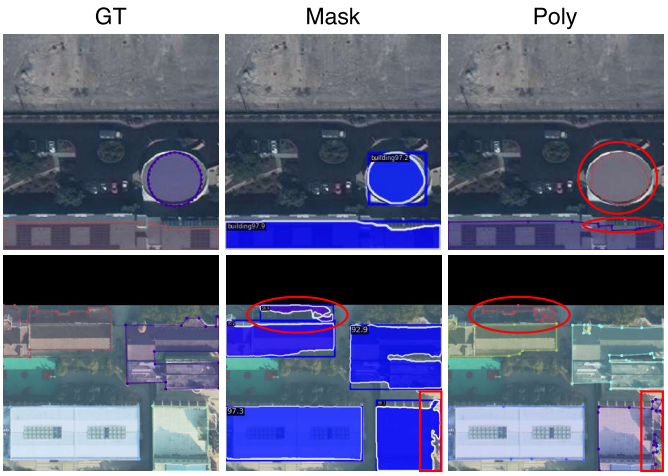


Fig. 8. Failure cases of the proposed GCP method. The first row is from the CrowdAI dataset, while the second row is from the WHU-Mix (vector) dataset.

performed asynchronously on CPUs, ensuring they do not introduce additional latency to the overall processing pipeline.

D. Failure Cases

We present two typical failure cases of the proposed GCP method on the CrowdAI and WHU-Mix (vector) datasets in Fig. 8. In the first row, GCP struggles with rounded building structures, as it tends to approximate building geometry using a piecewise linear representation, resulting in fewer vertices and less accurate outlines. The second row illustrates that the predicted building polygons are dependent on the predicted binary building masks. If the masks fail to capture the rough building geometries accurately, the polygon extraction process is also likely to fail.

A potential solution for better polygonization of rounded buildings is to adaptively adjust the regularization coefficient λ based on building type, allowing more vertices for curved structures. For the second issue, GCP's performance could be improved by integrating a more powerful instance segmentation model.

V. Conclusion

In this article, we propose a novel polygonal building mapping method called GCP, aimed at addressing the challenges of converting building mask representations into precise polygon representations. We introduce a polyline refinement module to enhance initial polygons sampled along the boundaries of building masks. This is followed by a global collinearity-based polyline simplification module, which processes the refined polylines to generate the final predicted polygons. In addition, we incorporate a global collinearity loss that is differentiable with respect to the refined polylines, enabling the entire polygonization pipeline to be trained end-to-end.

Extensive experiments demonstrate that the proposed method delivers superior performance. Furthermore, the polyline simplification module, when applied independently, outperforms traditional methods such as the DP algorithm, extending its potential for broader applications.

References

[1] X. Xiang Zhu, Q. Li, Y. Shi, Y. Wang, A. Stewart, and J. Prexl, "GlobalBuildingMap—Unveiling the mystery of global buildings," 2024, *arXiv:2404.13911*.

[2] C. Robinson, F. Hohman, and B. Dilkina, "A deep learning approach for population estimation from satellite imagery," in *Proc. 1st ACM SIGSPATIAL Workshop Geospatial Humanities*, Nov. 2017, pp. 47–54.

[3] J. Rentschler, M. Salhab, and B. A. Jafino, "Flood exposure and poverty in 188 countries," *Nature Commun.*, vol. 13, no. 1, p. 3527, Jun. 2022.

[4] J. Guo, D. Hong, and X. X. Zhu, "High-resolution satellite images reveal the prevalent positive indirect impact of urbanization on urban tree canopy coverage in South America," *Landscape Urban Planning*, vol. 247, Jul. 2024, Art. no. 105076.

[5] Y. Jiang, S. Yin, K. Li, H. Luo, and O. Kaynak, "Industrial applications of digital twins," *Phil. Trans. Roy. Soc. A*, vol. 379, no. 2207, 2021, Art. no. 20200360.

[6] S. Chen, L. Mou, Q. Li, Y. Sun, and X. X. Zhu, "Mask-height R-CNN: An end-to-end network for 3D building reconstruction from monocular remote sensing imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 1202–1205.

[7] B. Xu, J. Xu, N. Xue, and G.-S. Xia, "HiSup: Accurate polygonal mapping of buildings in satellite imagery with hierarchical supervision," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 284–296, Apr. 2023.

[8] N. Girard, D. Smirnov, J. Solomon, and Y. Tarabalka, "Polygonal building extraction by frame field learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 5891–5900.

[9] W. E. Lorensen and H. E. Cline, "Marching cubes: A high resolution 3D surface construction algorithm," in *Proc. 14th Annu. Conf. Comput. Graph. Interact. Techn.*, Aug. 1987, pp. 163–169.

[10] S. Suzuki and K. Abe, "Topological structural analysis of digitized binary images by border following," *Comput. Vis., Graph., Image Process.*, vol. 29, no. 3, p. 396, Mar. 1985.

[11] D. H. DOUGLAS and T. K. PEUCKER, "Algorithms for the reduction of the number of points required to represent a digitized line or its caricature," *Cartographica*, vol. 10, no. 2, pp. 112–122, Dec. 1973.

[12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 60, 2017, pp. 84–90.

[13] Y. Hu, Z. Wang, Z. Huang, and Y. Liu, "PolyBuilding: Polygon transformer for building extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 15–27, May 2023.

[14] M. Zhang, Q. Liu, and Y. Wang, "HiT: Building mapping with hierarchical transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5606316.

[15] R. Bellman, "Dynamic programming," *Science*, vol. 153, no. 3731, pp. 34–37, 1957.

[16] K. Zhao, J. Kang, J. Jung, and G. Sohn, "Building extraction from satellite images using mask R-CNN with building boundary regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 242–2424.

[17] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[18] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using CNN and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020.

[19] M. Li, F. Lafarge, and R. Marlet, "Approximating shapes in images with low-complexity polygons," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8630–8638.

[20] J.-P. Bauchet and F. Lafarge, "KIPPI: KInetic polygonal partitioning of images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3146–3154.

[21] M. Kass, A. Witkin, and D. Terzopoulos, "Snakes: Active contour models," *Int. J. Comput. Vis.*, vol. 1, no. 4, pp. 321–331, Jan. 1988.

[22] S. Wei, T. Zhang, S. Ji, M. Luo, and J. Gong, "BuildMapper: A fully learnable framework for vectorized building contour extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 87–104, Mar. 2023.

[23] S. Zorzi, S. Bazrafkan, S. Habenschuss, and F. Fraundorfer, "PolyWorld: Polygonal building extraction with graph neural networks in satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1848–1857.

[24] S. Zorzi and F. Fraundorfer, "Re: PolyWorld—A graph neural network for polygonal scene parsing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 16762–16771.

[25] S. Wei, T. Zhang, D. Yu, S. Ji, Y. Zhang, and J. Gong, "From lines to polygons: Polygonal building contour extraction from high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 213–232, Mar. 2024.
[26] J. Hosang, R. Benenson, and B. Schiele, "Learning non-maximum suppression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4507–4515.
[27] L. R. Medsker and L. Jain, "Recurrent neural networks," *Des. Appl.*, vol. 5, nos. 64–67, p. 2, 2001.
[28] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[29] Z. Li, J. D. Wegner, and A. Lucchi, "Topological map extraction from overhead images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1715–1724.
[30] W. Huang, H. Tang, and P. Xu, "OEC-RNN: Object-oriented delineation of rooftops with edges and corners using the recurrent neural network from the aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604912.
[31] X. Shi, Z. Chen, H. Wang, D. Yeung, W. K. Wong, and W. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.
[32] T. Zhang, S. Wei, Y. Zhou, M. Luo, W. Yu, and S. Ji, "P2PFormer: A primitive-to-polygon method for regular building contour extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4414012.
[33] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[34] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*Cham, Switzerland: Springer, 2020, pp. 213–229.
[35] B. Cheng, A. G. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. NIPS*, Dec. 2021, pp. 17864–17875.
[36] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.
[37] F. Li et al., "Mask DINO: Towards a unified transformer-based framework for object detection and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3041–3050.
[38] H. W. Kuhn, "The Hungarian method for the assignment problem," *Nav. Res. Logistics (NRL)*, vol. 52, no. 1, pp. 7–21, Feb. 2005.
[39] crowdAI. (2023). *Mapping Challenge*. Accessed: Sep. 28, 2023. [Online]. Available: https://www.crowdai.org/challenges/mapping-challenge
[40] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.
[41] Y. Kumar Adimoolam, B. Chatterjee, C. Poullis, and M. Averkiou, "Efficient deduplication and leakage detection in large scale image datasets with a focus on the CrowdAI mapping challenge dataset," 2023, *arXiv:2304.02296*.
[42] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[43] OpenAI. (2023). *2018 Open AI Tanzania Building Footprint Segmentation Challenge*. Accessed: Sep. 28, 2023. [Online]. Available: https://competitions.codalab.org/competitions/20100
[44] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.
[45] C. Wang, J. Chen, Y. Meng, Y. Deng, K. Li, and Y. Kong, "SAMPolyBuild: Adapting the segment anything model for polygonal building extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 707–720, Dec. 2024.
[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[47] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[48] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.



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